# NERON DESINGULARIZATION: AN UNRAMIFIED EXTENSION OF DVRS IS A FILTERED COLIMIT OF SMOOTH ALGEBRAS

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*Editorial note.* Input lemma-neron-desingularization is a published entry (tag 0BJ6) used as a standing input by its display number; not proven on the author's behalf.

*Allowed standard prerequisites (editorial).*

- Colimit criterion for ind-smooth maps. Algebra, lemma when-colimit (cross-chapter, named). Reduces to factoring every finitely presented A -> Lambda through a smooth algebra.
- Finite blowup sequence desingularizes. lemma-neron-desingularization (same chapter, PUBLISHED 0BJ6). After finitely many Neron blowups R -> A becomes smooth at p.
- Smoothness at the generic point of a separable domain map. Algebra, lemma smooth-at-generic-point (cross-chapter, named). R -> A is smooth at q = (0) after replacing A by its image in Lambda.

*Source reference.* Original reference 07ND: focused-seeds/geometry/S9-smo-neron-colimit/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 07C3: focused-seeds/geometry/S9-smo-neron-colimit/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 0BJ6: sources/stacks-smoothing/pinned-original-source.tex, source lines 1258–1258. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 0BJ2: sources/stacks-smoothing/pinned-original-source.tex, source lines 1085–1085. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

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*Date*: stacks-project:smoothing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Lemma 0.1.** *Let $R \subset \Lambda$ be an extension of discrete valuation rings which has ramification index 1 and induces a separable extension of residue fields and of fraction fields. Then $\Lambda$ is a filtered colimit of smooth $R$-algebras.*

*Proof.* By Algebra, Lemma 07C3 it suffices to show that any $R \to A \to \Lambda$ as in Situation 0BJ2 can be factored as $A \to B \to \Lambda$ with $B$ a smooth $R$-algebra. After replacing $A$ by its image in $\Lambda$ we may assume that $A$ is a domain whose fraction field $K$ is a subfield of the fraction field of $\Lambda$. In particular, $A$ is separable over the fraction field of $R$ by our assumptions. Then $R \to A$ is smooth at $\mathfrak{q} = (0)$ by Algebra, Lemma 07ND. After a finite number of Néron blowups, we may assume $R \to A$ is smooth at $\mathfrak{p}$, see Lemma 0BJ6. Then, after replacing $A$ by a localization at an element $a \in A$, $a \notin \mathfrak{p}$ it becomes smooth over $R$ and the lemma is proved. □